

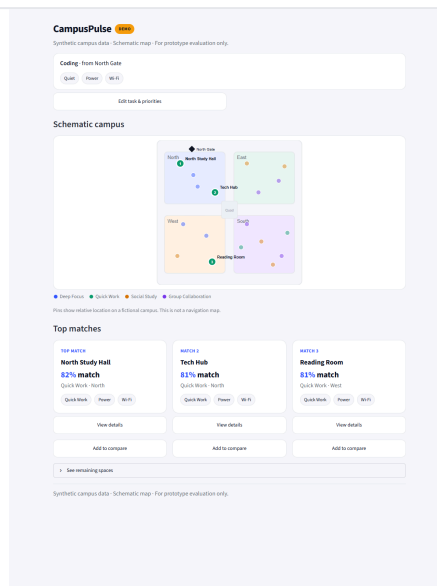
CampusPulse

SYNTHETIC / ILLUSTRATIVE

Schematic campus · 18 fictional study spaces · Streamlit prototype

How might task-aware data help students compare study-space trade-offs and choose a space that fits what they need to do?

CampusPulse is a lightweight decision prototype. It does not navigate a real campus and it does not use observed student surveys. It asks *what are you doing today?*, then makes the trade-offs visible enough to choose with confidence.



Synthetic campus data · Schematic map · For prototype evaluation only.

Overall rating cannot answer “where should I study today?”

A library can be excellent for exam revision and a poor fit for group discussion. A café can be comfortable and still fail a two-hour coding session if power and quiet are missing. Generic maps and review platforms collapse those differences into distance, opening hours, and a single score.

A good study space is not universally good — it is good for a particular task, at a particular moment, for a particular set of priorities.

What users actually need to compare

- Quiet vs conversation
- Power and Wi-Fi vs atmosphere
- Room to focus vs a short walk
- Why a recommendation appeared, and what it sacrifices

CampusPulse treats that comparison as the product, not a dashboard of extra charts.

A synthetic prototype, on purpose

The original research framing assumed field collection: 15–25 real campus places, 50–100 student responses, and a geographic map. Those inputs were not available. Continuing to describe the work as “student-generated data” would have been dishonest.

Original plan	Constraint	Prototype substitute
Real campus locations	No field access	18 labelled fictional spaces
Student survey ratings	No recruited sample	Archetype centres + seed-42 jitter
Folium / real coordinates	No geography	0–100 schematic x/y + inline SVG
Live community pulse	No backend	Session-only demo pulse
Usability study	No participants	Functional tests + illustrative benchmark

Every dataset and evaluation artefact in this case study is synthetic / illustrative. None of it is a human-subject finding.

18 fictional spaces from four archetypes

Spaces inherit a design centre — Deep Focus, Quick Work, Social Study, or Group Collaboration — then receive a small deterministic perturbation (`numpy` RNG seed 42). Task-fit scores (coding, reading, writing, group work, exam revision) are derived from attributes, not invented independently.

ID	Name	Zone	Precomputed cluster
S01	Library 4F	North	Deep Focus
S02	Library 2F	North	Deep Focus
S03	North Study Hall	North	Quick Work
S04	Tech Hub	North	Quick Work
S05	East Cafe	East	Social Study
S06	Student Centre	East	Group Collaboration
S07	Innovation Lounge	East	Group Collaboration
S08	Commons Cafe	East	Social Study
S09	Silent Annex	West	Deep Focus
S10	Garden Room	West	Deep Focus
S11	West Lounge	West	Social Study
S12	Reading Room	West	Quick Work
S13	South Project Room	South	Group Collaboration
S14	Residence Lounge	South	Social Study
S15	Late Lab	South	Quick Work
S16	Commons Booths	South	Quick Work
S17	Atrium Tables	South	Social Study
S18	Seminar Hub	South	Group Collaboration

Coordinates are schematic only. They support spatial scanning in the UI, not walking directions.

K-Means types and PCA — analysis, not the map

Clustering is a precompute step so the 18 spaces receive interpretable type labels for the map legend. It is not a live recommender and is never re-run on a Streamlit request. Implementation is NumPy-only (no scikit-learn): z-score standardisation, k=4 K-Means, PCA via SVD.

- Cluster sizes: Deep Focus 4 · Quick Work 5 · Social Study 5 · Group Collaboration 4
- PCA explained variance: PC1 47.4% · PC2 38.2%



Figure: 2D PCA of standardised attributes. Axes are statistical, not campus geography. Source: synthetic `locations.csv`, seed 42.

Task-first, decision over dashboard, explain the trade-off

Users should not tune an algorithm. They should say what they are doing, nudge a few plain-language priorities, and see a shortlist they can defend.

- **Task-first.** Discover starts with Coding / Reading / Writing / Group work / Exam revision.
- **Decision over dashboard.** The map is for spatial sense. Top 3 cards do the deciding.
- **Explain the trade-off.** Two reasons plus one cost, never raw weights.
- **Schematic, not geographic.** No tiles, no fake walking minutes.
- **Synthetic, not observed.** The DEMO badge and disclaimer stay visible in product mode.

Priority chips (max three) only boost a preset. The defaults already encode the task; the chips are a nudge, not a control panel.

Transparent score, not a black box

Each task has a design preset whose weights sum to 1. Choosing a priority adds $+0.08$ before renormalisation. Ratings scale from 1–5 into 0–1. Distance is relative effort on the 0–100 canvas.

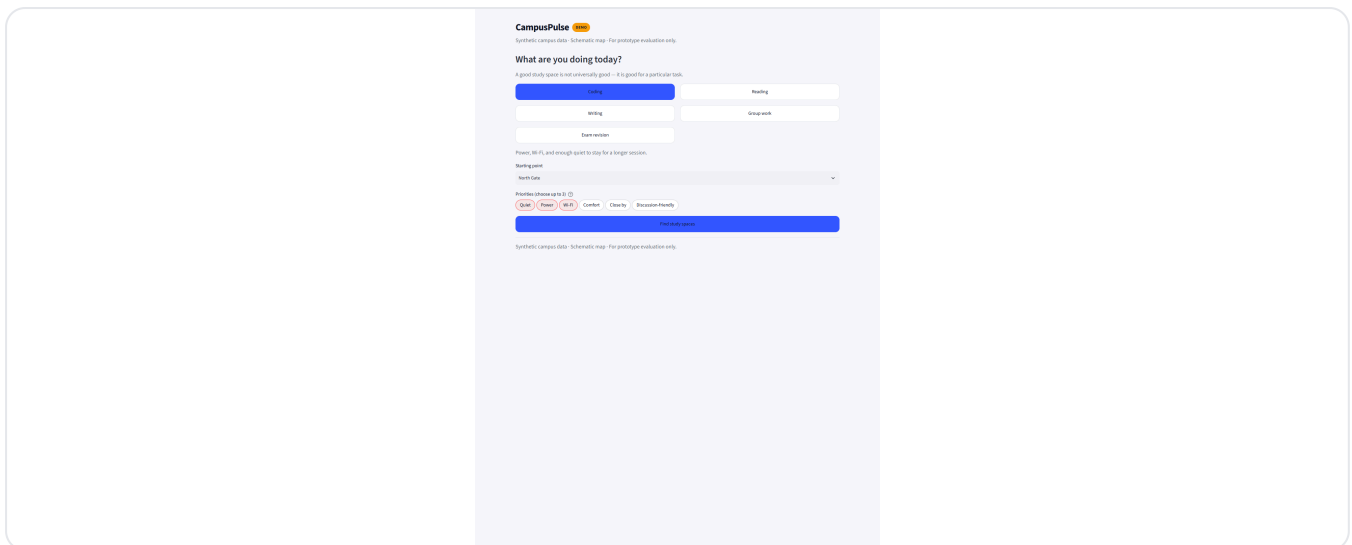
$$\text{Match} = 100 \times (0.70 \times \text{attribute score} + 0.30 \times \text{task fit})$$

The 70/30 split keeps the interface explainable while still respecting task-fit. For the default Coding path (North Gate · Quiet / Power / Wi-Fi) the Top 3 is stable:

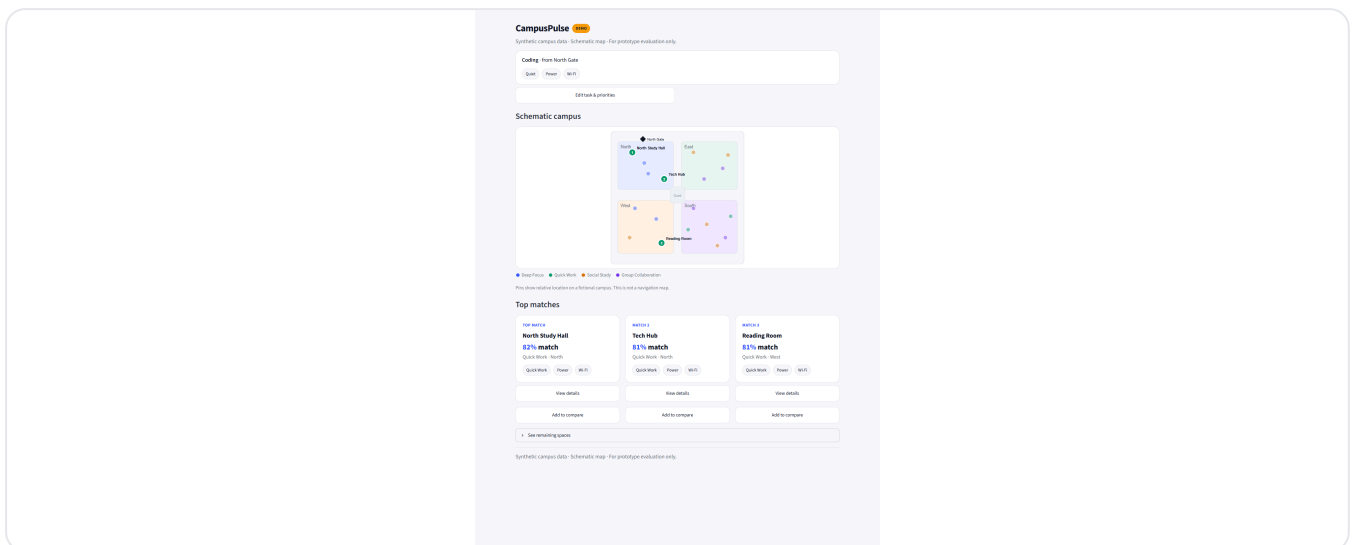
Rank	Space	Type	Match
1	North Study Hall	Quick Work	82%
2	Tech Hub	Quick Work	81%
3	Reading Room	Quick Work	81%

Group work with Discussion-friendly / Comfort is dominated by Group Collaboration spaces. Exam revision with Quiet / Comfort is dominated by Deep Focus. Writing + Close by lifts spaces near the chosen start. These are regression checks on synthetic data, not field outcomes.

Ask the task, then show the campus

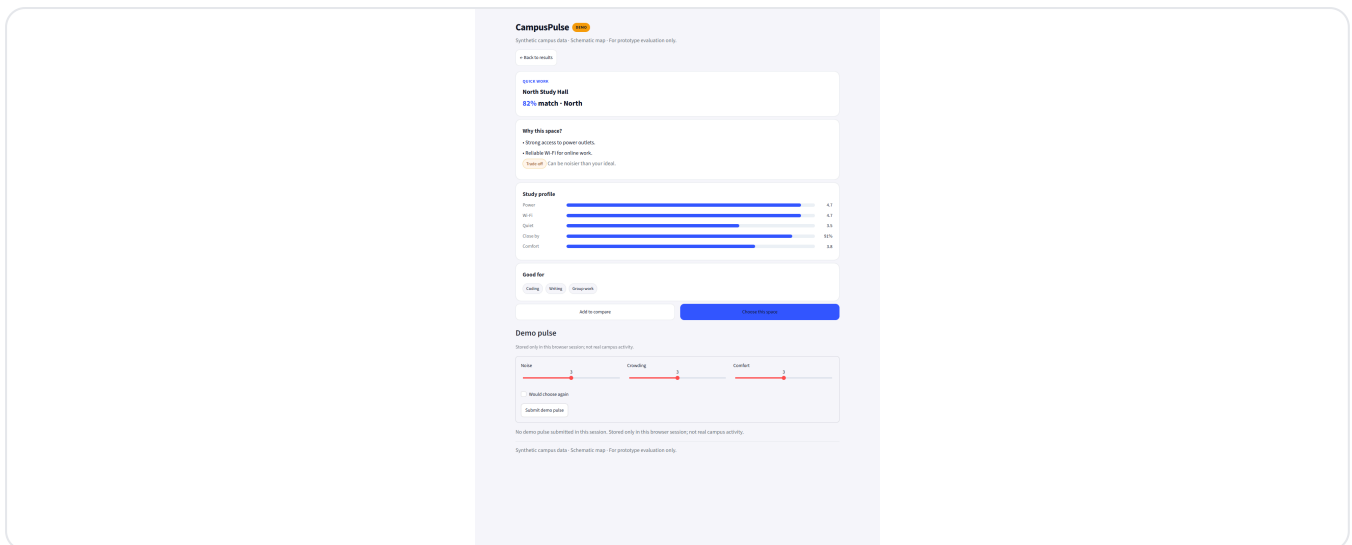


Discover: task cards, schematic start point, at most three priorities. The primary action is Find study spaces.



Results: numbered pins on a fictional campus, Top 3 cards, remaining spaces folded away. Not a navigation map.

Reasons first, then the profile



North Study Hall for Coding is recommended mainly for power and Wi-Fi. The stated trade-off is noise. Horizontal bars replace a radar chart so differences stay readable without extra chart libraries.

Demo pulse (noise, crowding, comfort, would choose again) is optional and session-only. The UI states: *Stored only in this browser session; not real campus activity*. There are no fake community counts.

Match score is a guide, not the answer

The screenshot shows the 'Compare' section of the CampusPulse application. At the top, it says 'CampusPulse' with a logo and a disclaimer: 'Synthetic campus data - Schematic map - For prototype evaluation only'. Below this is a 'Back to results' link. A blue banner states: 'North Study Hall leads overall, but Tech Hall is a close alternative.' Below the banner, a table compares two schools across six attributes. The 'Match score' is 82% for North Study Hall and 81% for Tech Hall. The table shows scores for various attributes, with the highest score in each row highlighted in green. Below the table, there are buttons to 'Remove from comparison' for both schools and 'View details' for each. At the bottom, the same disclaimer is repeated.

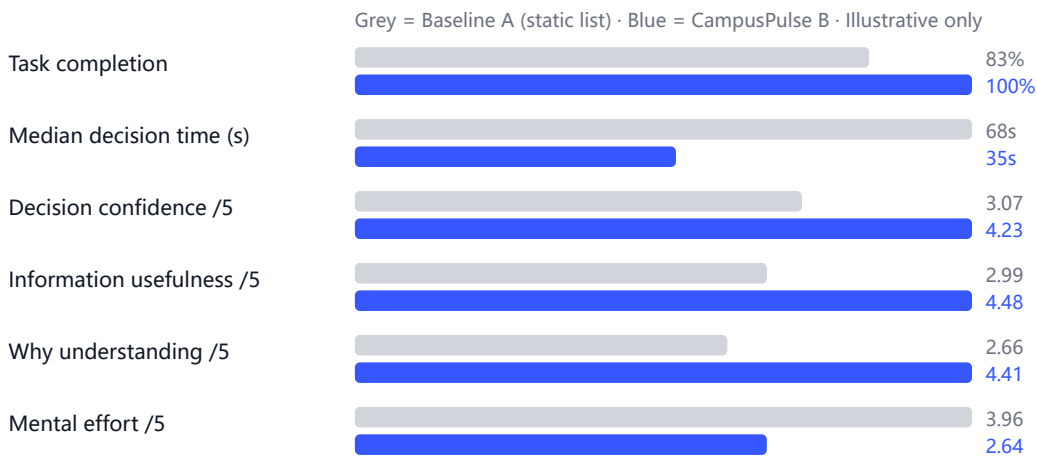
Attribute	North Study Hall	Tech Hall
Match score	82%	81%
Power	4.7	4.8
Art	4.7	4.8
Spout	3.5	3.6
Class by	80%	78%
Location	3.8	3.9
Water to house	3.1	3.2

Compare holds two or three spaces. A deterministic hint names a clear leader when the gap is at least 8 points, otherwise it names a close alternative. The table shows the six attributes that matter most for the current weights, with the stronger cell marked. Users can still disagree with the score — that is the point of showing the rows.

Functional tests, plus an illustrative benchmark

`pytest` covers data bounds, deterministic ranking, priority boosts, distance, explanations (2 reasons + 1 trade-off), compare cap, session-only pulse, and the four task scenarios. Streamlit AppTest checks that product mode keeps the synthetic disclaimer visible. No browser automation is required for the test suite.

Not a user study. The chart below is generated from `research/synthetic_evaluation.csv` (`data_type=synthetic_illustrative` , seed 2026). It defines layout and acceptance targets. It must not be captioned as n = 12 participants.



Source: `synthetic_evaluation.csv` · 12 simulated profiles × 3 tasks × 2 conditions · illustrative generator centres from the build spec.

What this prototype can and cannot claim

CampusPulse shows that task, priorities, and attributes can be turned into a short, explainable comparison without a real map or a live community. That is a design and engineering argument, not an efficacy result.

Limits

- Ratings come from designed archetypes, so clusters partly recover the structure that was put in.
- Distance is cartoon walking effort.
- Session pulse cannot stand in for campus activity.
- The illustrative benchmark must stay labelled. It is not evidence that decision time or confidence improved for real students.

If a real study becomes possible

A formative test with 6–8 students, three tasks, and a static-list control would be enough to find usability issues. It should be reported separately — not by rewriting these synthetic numbers as participants.

Engineering stack. Streamlit + Pandas + NumPy. Offline K-Means / PCA. Inline SVG map. No database, no tiles, no ML API.

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